

Module 6: Genetic Change

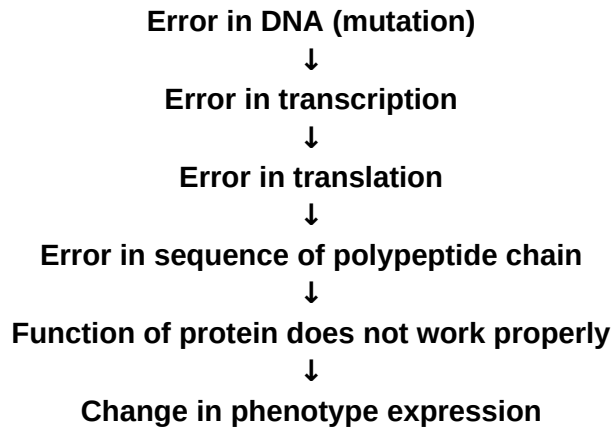
Mutation - How does mutation introduce new alleles into a population?

Allele: different forms in which one gene can be expressed

Mutations: permanent change to the DNA sequence of an organism. They can occur anywhere in a person's DNA sequence; protein coding gene, RNA gene, non-coding DNA.

Why are mutations important?

- Mutations create variations in phenotype and these variations can be inherited. This variation increases the chance of a population's survival during environmental change. (the only way to introduce new alleles into a population)
- Without mutations, natural selection cannot occur. Variations that help an organism survive & reproduce are passed to the next generation



The Criteria for Mutations:

1. **The origin of the mutation:** spontaneous or induced mutation (mutagen)
2. **The amount of genetic material changed:** point mutation or chromosomal mutation
3. **The effect of the mutation on DNA:**
 - Point mutations: substitution, insertion, deletion (frameshift)
 - Chromosomal mutations: deletions, insertions, translocations, duplications
4. **The effect of the mutation on phenotype:** silent, missense, nonsense
5. **Heritability of mutation:** somatic or germline

- explain how a range of mutagens operate, including but not limited to:

Mutagens

- Agents that lead to mutations
- Mutations can occur due to exposure to mutagens

Spontaneous Mutations

- Happens spontaneously / unpredictable
- Can occur when mistakes are made during DNA replication or transcription
- Not frequent & occur naturally during a person's lifetime

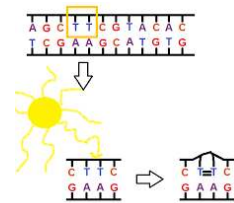
– electromagnetic radiation sources

Ionising Radiation - X Rays, Gamma Rays

Radiation can break chemical bonds causing damage to DNA. Multiple fractures in the DNA can overwhelm enzymes causing mistakes to be made and mutations to occur.

UV Rays

Ultraviolet radiation from the sun can promote a chemical reaction in the DNA, causing certain bases next to each other to fuse together e.g. two thymine bases. These thymine dimers cause a bulge in the DNA and can change or disrupt DNA expression, preventing normal transcription and replication



– chemicals

Intercalating agents

Some chemicals insert themselves between bases of a DNA sequence. This insertion causes the DNA structure to stretch, “fooling” DNA polymerase into inserting an extra base opposite the intercalating molecule causing frameshift

Base analogous

These chemicals are similar in physical structure to nitrogenous bases. During replication these molecules may be incorporated instead of nitrogenous bases and hence a mutation

Reacting Chemicals

These chemical mutagens react with the nitrogenous bases and chemically modify the DNA causing mutation.

Example: Carcinogen is a chemical found in cigarettes that causes damage to lung cells.

– naturally occurring mutagens

Biological Mutagens

Metabolism processes result in the release of naturally occurring chemicals such as free radicals. Free radicals cause oxidation of DNA, breaking DNA strands, which interferes with DNA replication and transcription.

Viruses

Viruses insert their genetic information into the chromosomes of the host cell, causing disruption to normal cell function e.g. HPV is an oncogenic (cancer causing) virus that causes cervical cancers.

- compare the causes, processes and effects of different types of mutation, including but not limited to:

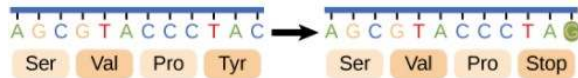
- **point mutation:** mutation changing one or very few nucleotide bases in a gene sequence e.g. **sickle cell anemia**

Base Substitution: one base is substituted for another.

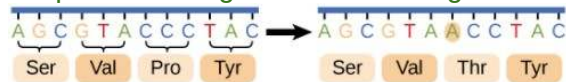
insertion or deletion: when an extra base is added or deleted -> only leads to frameshift

Effect on proteins

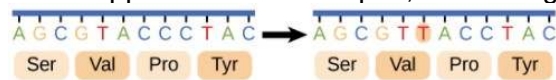
Nonsense: changing an amino acid to a stop codon, cutting the protein short. This resulting protein is normally non-functional



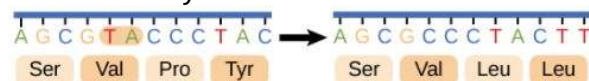
Missense: a point mutation that results in 1 amino acid change. E.g. sickle cell anaemia occurs when the triplet CTC changes to CAC causing glutamate to be swapped with valine, altering the shape of haemoglobin and causing sickle cell shaped blood cells.



Silent: a change in the DNA sequence that does not cause a change in amino acids. This is due to the swapped base in the triplet, still coding for the same amino acid as the old one.

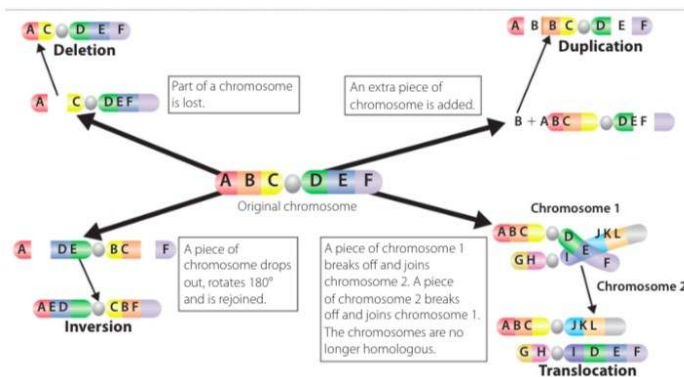


Frameshift mutation: an insertion or deletion shifts the entire reading frame of RNA, leading to the creation of a whole sequence of incorrect amino acids and production of a non-functional protein. Since the transcription of DNA is read in codons of three bases, a deletion or insertion of three bases will only affect one amino acid.



- **chromosomal mutation:** large scale mutations that affect the overall **structure** of a chromosome or the entire **number** of chromosomes in a cell are altered.

Changes to the Structure



Deletion: a section of DNA is removed and not replaced, causing a reduction in the number of genes in a chromosome.

Insertion/Duplication: a portion of DNA is duplicated then inserted, increasing the number of genes on a chromosome.

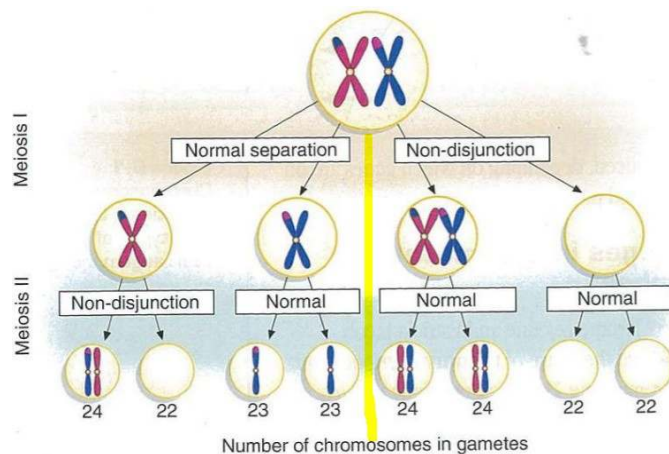
Inversion: a section of DNA is removed, turned 180 and then reinserted into the chromosome so that the bases are in reverse order.

Translocation: a section of DNA is moved from one chromosome to another.

Changes to Chromosome number (Aneuploidy)

- Caused by nondisjunction during meiosis or mitosis
- Nondisjunction is the failure of homologous chromosomes to separate during cell division (Meiosis1 - homologous chromosomes separate, Meiosis 2 - chromatids separate)
- Outcome are cells with extra or missing chromosomes

Trisomy: an extra copy of a chromosome present. Can lead to shortened life span or development problems.

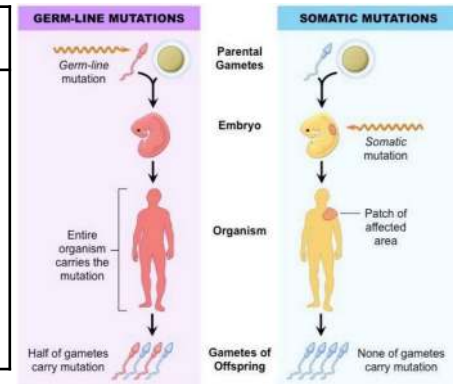


Example

Trisomy 21 means there are three copies of chromosome 21, resulting in **down syndrome**.

- distinguish between somatic mutations and germ-line mutations and their effect on an organism

Somatic Mutation	Germline Mutation
- Mutation in a body cell - Not heritable - Occurs throughout a person's lifetime - Present only at site of mutation - Mutated cell can divide, growing tissue with this error (cancer)	- Mutation in a sex/gamete cell (can be passed onto offspring) - Heritable - Can be inherited from birth - Mutation will occur in all cells of the offspring's body as original cell was already mutated



- assess the significance of 'coding' and 'non-coding' DNA segments in the process of mutation

The DNA sequences which encode for proteins are known as coding DNA. The sequences which do not encode for proteins are known as noncoding DNA.

	Segment	Function	Significance of a mutation?
Coding	Exons (gene)	Codes for proteins that carry out biological functions.	Can directly impact the sequence of amino acids and therefore affect the functioning of proteins. Changed DNA code = changed mRNA = mutated protein
Non-coding	Enhancer/ Silencers/ regulators	Enhancers turn on or enhance the expression of a gene. Silencers turn off or slow the expression of a gene.	Can cause too much or too little production of proteins or the wrong protein to be produced if turned on. E.g. diabetics
	Promoters	Tell the enzyme RNA polymerase where to bind RNA to DNA which begins the transcription.	Mutation can result in polypeptide chains not being fully transcribed, causing deficient proteins
	Introns (gene)	Code for their own removal during mRNA splicing (end of transcription)	Have the potential to affect splicing of DNA (mutation in introns causes removal of exons = deletion).

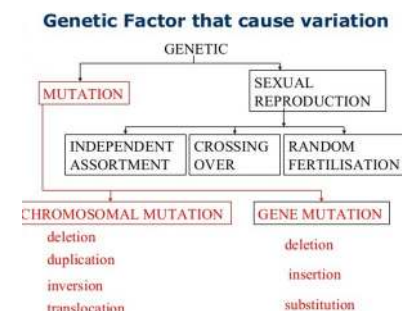
- investigate the causes of genetic variation relating to the processes of fertilisation, meiosis and mutation

Genetic Variation occurs due to:

Mutations: only source of new alleles within a population (germline mutations).

Sexual Reproduction: responsible for the combination of alleles.

- **Random Fertilisation:** Recombination of haploid cells during fertilisation is random, introducing different allele combinations
- **Meiosis:**
Law of segregation: homologous chromosomes (anaphase 1) and sister chromatids (anaphase 2) must separate



Independent assortment: chromosomes separate into daughter cells independently of one another (as in not all father chromosomes go to one cell and not all mother chromosomes go to the other cell)

Crossing over: new combinations of alleles so each gamete is unique

Genetic variation can be beneficial for the survival of the population as it gives some individuals an advantage over the rest of the population. Genetic variations can also be negative and lead to diseases or they can have no effect. Genetic variations are not as beneficial to a particular individual but are more helpful to the survival of the population if there are unexpected changes in their environment.

- evaluate the effect of mutation, gene flow and genetic drift on the gene pool of populations

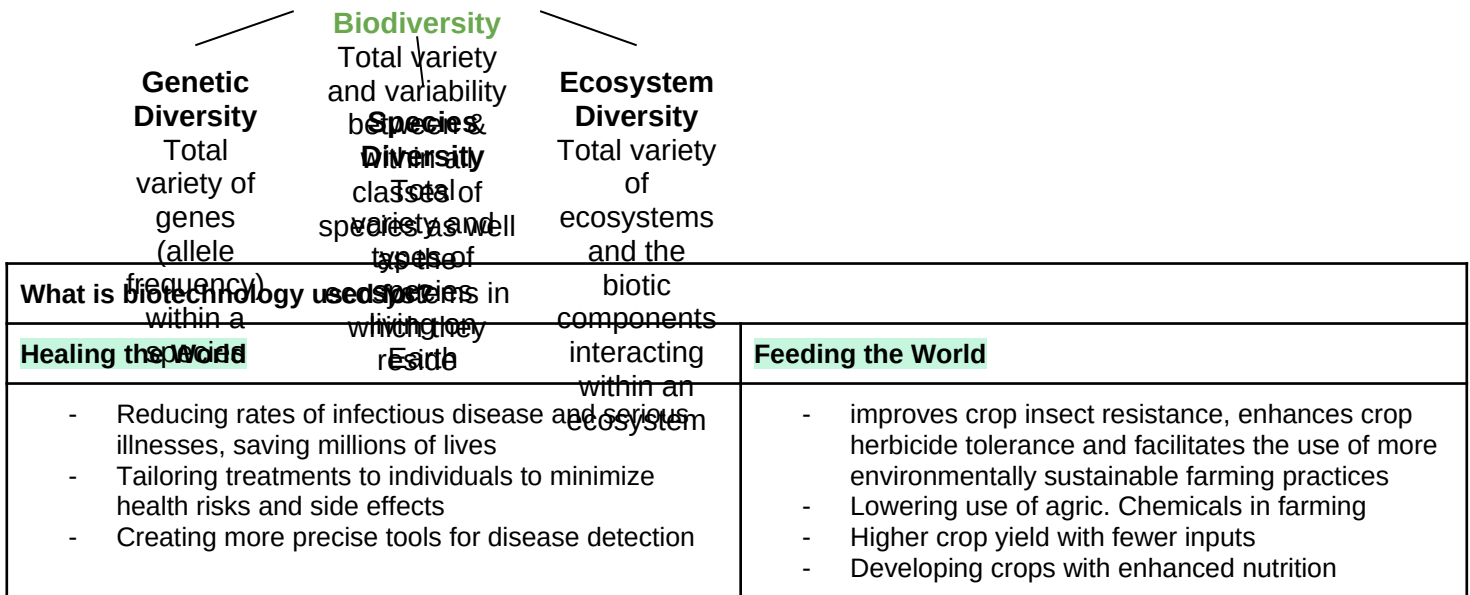
Mechanisms for changes in frequency of a gene pool:

Mechanism	Definition	Example	Effect on Gene Pool
Natural Selection	Selective pressures such as predators and mating result in certain characteristics increasing an individual's chance of survival. Therefore unfavourable traits are not passed to the next generation, reducing frequency of this allele in the gene pool		
Mutations	Direct change in the DNA of an individual, introducing a new allele into a population. The only way new alleles can be created within a population.	Introduction of new green frogs that are better suited to the environment, resulting in greater frequency of green frogs in the gene pool over time.	Increases number of alleles in a population
Genetic Drift (chance)	A random change in the environment causing random alleles in the population to be reduced in frequency. Different to natural selection as it is not linked to adaptive advantage to environmental change as it is CHANCE of who survives. Small populations are greatly affected by genetic drift.	Founder Effect: a small group of individuals are separated from the main population and become the founding members of a new population. This new population has different allele frequencies to the original population. (Armish have 6 fingers) Bottleneck Effect: lots of a population died due to random environmental change and by chance certain individuals survived (within this decreased population there is decreased genetic variation)	Decreases number of alleles in a population
Gene Flow (migration)	Movement of genes between populations. (each population introduces more alleles into the other population), increasing alleles present in gene pools. (migration) Alleles may be lost due to emigration (individuals that migrate take a specific allele out of the frequency) Gene flow can also result in introduction of a harmful mutation to a new population (disease). Lack of gene flow results in speciation.	A blonde-haired, blue-eyed family from Sweden moves to India where the children grow up, marry Indians and produce offspring who now have the blonde-haired, blue-eyed alleles	Increase and decrease the number of alleles in a population.

Evolution: changes in characteristics of a species overtime

- When the gene pool of a population changes, evolution occurs.

Biotechnology - How do genetic techniques affect Earth's biodiversity?



- investigate the uses and applications of biotechnology (past, present and future), including:
 - analysing the social implications and ethical uses of **biotechnology (not changes to embryo material)**, including plant and animal examples & evaluating the potential benefits for society of research using **genetic technologies (changing the DNA - only GMO, CRISPR, and cloning)**
 - evaluating the changes to the Earth's biodiversity due to genetic techniques

Biotechnology: any commercial exploitation of biological processes. Can be grouped in 4 categories: medicine, industry, environment, agriculture.

Genetic Technology	Social implications	Ethical implications
Selective Breeding, artificial pollination & hybridisation To get desired combination of favourable characteristics past	- Increase in yield production (pesticides, size, resistance, faster growth) = less land needed E.g. Corn selectively bred to have a bigger kernel. Mules were bred from a horse and a donkey. Mules therefore aided in transport as they were strong and had endurance E.g. Labrador and Poodle - characteristic for a family friendly dog that is hypoallergenic - Pest resistance can reduce pesticide runoff that can harm surrounding environment - Increase in health benefits of food - Higher profits - Faster process of reproduction so endangered species can be encouraged - Less environmental damage because of more efficient food production - Susceptibility to extinction - Inbreeding leads to diseases - Reduces biodiversity (consistent selection of the same traits)	- Exaggerated features can negatively affect the health of an animal - Is increase in global food and increased nutritional value being used to support developing countries or making an obese population more obese?
Fermentation Cheese, wine, beer, bread production. Bread was fermented by the biological organism yeast present in the	- use of natural microorganisms → a natural process used to make food for survival - Improve nutrition/yield of crops → reduces mass starvation and poverty	- people argue that microorganisms are unnatural for people to consume

air. Enzymes are added to milk to speed up the curdling process. past	- natural milk (without sterilization technology) is dangerous and illegal to sell (in the US) → sterilization is unnatural. Many have argued that raw milk is safer & healthier than pasteurized milk cultures. - humans need microbial life to survive, so fermentation gives necessary nutrients - opens new jobs	
Vaccines Contain weak/inactive antigens that trigger the immune system to resist it in the future. Newer vaccines contain blueprints for antigens rather than the antigen itself so the immune system knows how to fight it in the future. Medicine & Antibiotics Many medicinal plants are used to make medicine. The chemical penicillin was found to be produced by a fungus that stops bacteria from growing. present	- Less burden on health care systems as people are healthier through use of plants for medical treatment (mould was found to treat infections - penicillin has antibacterial properties) - Prevent disease before outbreak occurs - Can eradicate diseases (e.g. smallpox) - Longer life spans - Evolution of resistant strains of pathogens resulting from inappropriate use of antibiotic has resulted in new strains of viruses - Changing the genomes of other organisms and introducing them to an ecosystem can lead to invasive behaviour and reduction in biodiversity - Identification of a plant than can be used for pharmaceutical purposes would raise the value of the material, resulting in exploitation and species loss	- Developing countries who often need these vaccines or medicine most cannot afford - Vaccines and antibiotics are often tested on animals which can harm their health or cause pain and distress - Deliberately breeding animals for research such as promoting tumor growth can harm animals
Artificial Insemination Taking the sperm from a male with desired traits and artificially inserting it into several females present	- Transport of sperm to other communities to increase genetic variation - Animal conservation to maintain biodiversity by encouraging reproduction (Southern white rhino was brought back from the brink of extinction) - Desired traits are passed onto offspring, increasing output / resources (meat, milk) - decreases genetic variation as certain traits are continuously selected, artificially increasing the allele frequency of these traits - susceptibility to extinction due to decreased variation - inbreeding leads to diseases - genetically modified crop varieties severely reduces the number of non-GMO varieties causing loss of traditional varieties & decrease in biodiversity - Possibility for GMOs to have unknown negative health side effects due to differences in nutritional content, allergic response, or undesired side effects such as toxicity, organ damage, or gene transfer	- Who has access to these genetic resources, can cause unfair sharing of benefits - Could be stated as an unnatural process - Artificial insemination can appear to violate animal rights, because they involve impregnating animals for human ends as if the animals were nothing more than human property, rather than treating the animals as being of value in themselves
Transgenic species (GMO) One or more DNA sequences from another species is introduced by artificial means. Animals usually are made transgenic by having a small sequence of foreign DNA injected into a fertilized egg or developing embryo. present	- Can manipulate food to provide greater nutritional value (addressing hunger in poor and developing countries - Golden Rice has high Vit. A content) - Increased yield, disease and pest resistance, weather resistance and enhanced flavor (you don't have to spray plants with chemicals = less damage to environment = healthier population) - Potential to have pharmaceutical uses (Tracy the sheep was the first transgenic mammal created - produced the human protein alpha1-antitrypsin in her milk (was a potential pharmaceutical for the treatments of cystic fibrosis and emphysema) - Transgenic organisms will become dominant as they were modified to increase survival, therefore causing extinction of natural organisms (genetic erosion) - Transgenic organisms introduced to an ecosystem can lead to invasive behaviour and loss of biodiversity (changes biodiversity of natural environment) - We don't know the long term effects of eating GMO foods (new allergens may be created e.g. GMO soybeans were using genes from brazil nut - people with allergies to brazil nut are allergic to soybeans) - Farmers in the developing work cannot compete in markets	- Who has access to these genetic resources, can cause unfair sharing of benefits - Can cause an organism to have harmful characteristics (e.g. Modern pigs have been bred to grow extra fast - some breeds now grow too fast for their hearts, causing discomfort when animals are too active) - Genetic engineering and selective breeding appear to violate animal rights, because they involve manipulating animals for human ends as if the animals were nothing more than human property, rather than treating the animals as being of value in themselves - Vegetarians or religious people may be concerned animal genes are used in plant products - Is adding human genes to animal genomes immoral - Negative environmental impact

	as they do not have access to GMO technology (social inequality)	
DNA sequencing Technology can sequence the entire DNA base sequence of your chromosomes to identify variants that may be disease causing. present	- Provide guidance on future treatment options - Increase awareness of susceptibility to a hereditary disease - allowing for preventative measures to be taken and early detection - Allows us to identify embryos with genetic diseases (BRCA1 or BRCA2 carriers) - Can be very expensive	- Can have negative psychological impacts on a patient who carries a harmful mutation such as a BRCA gene - Certain individuals do not want to know their risk to the disease - Can cause descrimination in health insurance - Privacy of health organisations having your DNA sequence - Ethical issue of where DNA sequencing stops as screening embryos could lead to selective breeding (designer babies)

– researching future directions of the use of biotechnology

Cloning Researchers remove a mature somatic cell, such as a skin cell, from an animal that they wish to copy. They then transfer the DNA of the donor animal's somatic cell into an egg cell that has had its own DNA / nucleus removed. future	- Can increase genetically 'perfect' animals or crops that provide the greatest produce (output, nutrition, quality, growth rate) - Cloning helps with regenerative medicine and stem cell treatment for burn victims, reconstructive surgeries etc. - Can help with plants or animals who cannot reproduce - Can be used as a source for organ replication for organ transplants - Decreases genetic diversity - Cloning an organism does not provide an exact replication as environmental factors affect phenotype expression - If animal A has the a disease/fault, all clones will have this fault	- Cloning goes against certain religions - Uncertain/unpredictable results: over 89% of animals and human cloning attempts have recorded a fail, which means that animal DNA is only exposed to risk in the whole process - Only the rich can afford - Human clones become a commodity to be bought and sold
Gene editing / Splicing Cutting out genes using restriction enzymes (Cas9) that cut DNA at a specific base. CRISPR-Cas9 Cas9 enzyme contains a guide DNA which is complementary to the targeted gene. The Cas9 DNA cutting enzyme is introduced to a cell and locates the DNA strand and cuts it. Then a replacement healthy DNA strand is replaced. future	- Gene editing could completely treat diseases promptly and not just the symptoms, reducing and eliminating future health consequences (by removing harmful DNA mutations) - Some gene editing tools are inexpensive compared to current therapies, benefiting individuals and the economy (CRISPR). - Gene editing is a recent technology that isn't fully researched, there are potential side effects such as cancer - Can reduce biodiversity (GM mosquitoes) - A gene drive can help improve "organism."	- The choice of being able to edit and design babies is a controversial topic - Can possibly cause future inequality in society. Gene-editing can create 'designer babies' with enhanced physical features, intelligence, or athleticism that may create superiority in society. - May be against religious beliefs - Can cause racial descrimination

Answering question: if a question asks for genetic technology you can't use all biotechnology

- Define biotechnology
- Give examples of a few biotechs
- What are multiple examples of benefits of these two biotechnologies

Genetic Technologies - Does artificial manipulation of DNA have the potential to change populations forever?

Genetic Technology: a range of activities concerned with understanding gene expression, taking advantage of natural genetic variation, modifying genes and transferring genes to new hosts. Gene technology is a modern branch of biotechnology that allows direct manipulation of DNA or removal of a gene, or the transfer of a gene from one species to another. E.g. transgenic organisms

- investigate the uses and advantages of current genetic technologies that induce genetic change

		Uses & Advantages
Reproductive technologies	Artificial Insemination (in uterus)	- Improves conservation efforts of endangered species - Allows couples experiencing infertility to fall pregnant - Can produce stronger more efficient produce through selecting desired traits to pass onto the next generation - Allows control over the time of breeding - Allows the storage of sperm and insemination into multiple individuals
	In Vitro Fertilisation (in lab)	- A process through which an egg is removed from the ovaries and fertilised by injecting a single sperm into the egg then transferred into the uterus - Higher fertility rates than artificial insemination - Can be used for conservation of animal species - Can help mothers struggling with contraception to enable pregnancy - IVF allows farmers to choose the healthiest animals to reproduce allowing for greater output and quality e.g cattle are bred for a high quality meat
	Artificial Pollination	- Plants with the ideal characteristics such as size and seed number can be used for reproduction to increase yield, also meaning less strain on the environment - Scientist have created medicine from plants - Higher yield = higher profit for farmers = helping developing countries
Cloning techniques	Gene Cloning	- Larger and more resilient food can be grown - Genes can be copied to allow for DNA sequencing - Transgenic E-coli can produce insulin for people with diabetes - <u>Create copies of DNA or segments of DNA to be inserted into animal embryos to produce transgenic organisms</u> - Can be used to grow replacement organs - Genes can be used to inserted into bacteria to create transgenic bacteria (adenovirus in gene therapy - virus delivers gene to affected tissue)
	Whole-organism cloning	- Can replicate animals or plants with ideal characteristics to maintain high output levels - Increase population numbers fast - Create security in the global food supply
Recombinant DNA technology (use these in exams)	Gene Sequencing	- A laboratory technique used to find the exact sequence of nucleotide bases in the sequence of a patient's DNA - Used to identify location and function of a gene that can be used in transgenic organisms - Used to identify DNA variants / mutations in the gene sequence that may suggest the cause of a disease e.g. if a person carries a BRCA1 or BRCA2 gene that increases the risk of breast cancer occurring - Knowledge of a mutation can guide doctors on treatment options and preventative measures for a disease such as gene therapy - Scientists can use sequence information to determine which stretches of DNA contain genes and which stretches carry regulatory instructions, turning genes on or off - Can suggest family members may also carry a genetic disease increasing awareness - A study has found the entire sequence of a healthy human and therefore it is easy to identify a mutation - New methods of gene sequence can now target specific chromosomes, therefore decreasing costs of procedure and time - Can be used in combination with pre-implantation screening to reduce diseases within a

		population
	Transgenesis	- Enables the transfer of a gene from one species to another - Could lead to larger yields, capable of crops growing under stressful environments, resistance to disease and pests, enhanced nutrition composition, faster growing and improved quality of e.g. silk - Can decrease spread of specific diseases amongst animals - Can create disease restraint medication
	Gene Therapy Normally involves taking a desired gene and inserting it into modified viruses that cannot cause a disease. This virus then delivers genetic material into a cell's nucleus causing the cell to express this gene.	- Can be used to repair / replace faulty genes in an attempt to cure a disease or improve the bodies ability to fight diseases - Can be done at embryonic stage to prevent the development of a hereditary disease - Can regulate gene expression - No donor or other organisms DNA is needed - holds promise for treating a wide range of diseases, such as cancer, cystic fibrosis, heart disease, diabetes, hemophilia and AIDS
	CRISPR CRISPR is one laboratory method of altering the DNA and can be used as a tool for gene therapy. It involves the cutting and removal of a gene	- Has the ability to edit a DNA sequence by cutting out or putting in nucleotide bases - Can correct genetic defects -> treating and preventing a disease -> - Can develop new plant varieties that are indistinguishable from those developed through traditional breeding methods - GM mosquitoes - Researchers have used CRISPR/Cas9 gene editing to target a specific gene tied to fertility in male mosquitoes. GM male mosquitoes carry a gene that causes offspring to prematurely die. They are released in large numbers to overwhelm existing male populations. This suppresses population growing as females mate with GM males, reducing reservoir size - Can reduce biodiversity as it is a food source in the food chain
	Genetically Engineered Vaccines RNA vaccines are created by inserting viral RNA into cells. This viral RNA codes for surface protein antigens. This stimulates the immune system response against the virus.	- Virus RNA that codes for antigen producing genes are injected into the DNA of human cells. These cells synthesis virus antigens that stimulate the immune system response - Alternatively RNA that codes for antigens is inserted into the plasmid of bacteria. Recombinant bacteria multiply and produce virus antigens. Virus antigens are extracted and used in the vaccine.

• compare the processes and outcomes of reproductive technologies, including but not limited to:

- artificial insemination

Male gametes are collected and inserted into the uterus of a female.

- Artificial pollination

Pollen is collected from the male part of a flower (stamen) and transferred to the female part of a flower (stigma).

Process		Outcome	
Similarity	Differences	Similarity	Differences
Both involve collection of male gametes which are then transferred	- artificial Insemination involves the transfer of sperm one female at a time - Sperm can be frozen and transported over longer periods of time	allow to produce offspring with desired characteristics but both lead to a	artificial pollination involves production of flowers plants while artificial

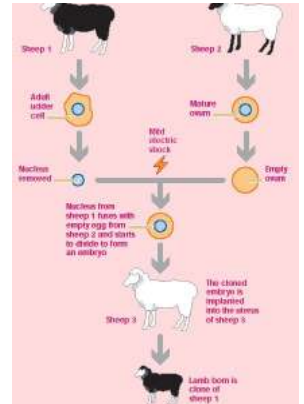
into/onto the female.	- artificial pollination requires the collection of pollen by milling flows which are then blown or sprayed onto multiple plants at a time - Also if you want to stop self pollination anthers have to be cut off and bags need to be placed over the flower to stop further cross pollination	reduction in genetic diversity.	insemination is used to produce animals
-----------------------	---	---------------------------------	---

- investigate and assess the effectiveness of cloning, including but not limited to:

- Whole Organism Cloning

Somatic cell nuclear transfer

1. A somatic cell has its nucleus removed from sheep 1
2. A mature egg is removed from sheep 2 and its nucleus is taken out
3. Nucleus from sheep 1 is inserted into egg from sheep 2
4. Electricity stimulates division of zygote to form an embryo
5. Embryo is implanted into a surrogate mother and will create an identical offspring to sheep 1

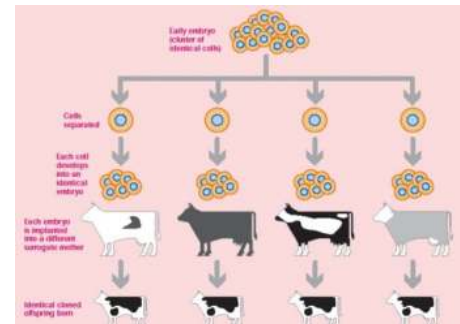


Artificial twinning

1. Early embryo (cluster of identical cells) is collected
2. Cells are separated and continue to divide forming separate embryos
3. Each embryo is implanted into a separate surrogate mother

Effectiveness: more effective than selective breeding in producing organisms with desired features but it can be more expensive

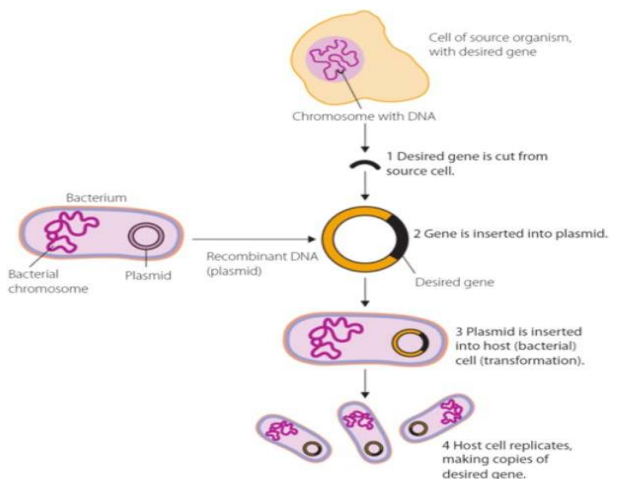
- gene cloning



Inserting a Gene into a bacteria

1. Cut out desired gene using restriction enzymes
2. Cut the bacterial plasmid with the same restriction enzymes
3. Use DNA ligase to paste gene into the plasmid
4. Using heat shock, Insert the plasmid into bacteria
5. When bacteria goes through binary fission, they make copies of the gene

E.g. cloned bacteria produce insulin which is harvested for patients with diabetes



PCR (Polymerase chain reaction)

1. A sequence of DNA sample is extracted
2. Denature DNA - Heat separates the double stranded DNA into single strands

3. Add free nucleotides, primer and polymerase
4. Polymerase adds free nucleotides to exposed DNA strands (starting at the primers) to make a copy of the DNA
5. This process is repeated making many copies of DNA

Effectiveness: allows us to copy genes that can be used in combination with other genetic technologies to make useful products for us.

- describe techniques and applications used in recombinant DNA technology, for example:

Recombinant DNA Technology: technology that combines two genes from different species
e.g. transgenic species / genetically modified species

- the development of transgenic organisms in agricultural and medical applications

Transgenic Bacteria

1. Identify the desired gene from an organism (e.g. insulin gene from human)
2. Isolate the gene by using restriction enzymes to cut it out
3. Make multiple copies of the gene using a PCR machine
4. Using the same restriction enzyme to cut multiple bacterial plasmids in E.coli
5. Use DNA ligase to paste the genes into the cut plasmid to make multiple **recombinant plasmids**
6. Add promoter and regulator sequence to the gene to enable it to be expressed
7. Use heat shock to insert the recombinant plasmid into the bacteria to create a **transgenic bacteria**

Use in medical field

If successful the bacteria is able to produce human insulin which is used in medication to treat people with type 1 diabetes

Transgenic Animals

1. Identify the desired gene from an organism (e.g. antithrombin gene from humans - stops clotting)
2. Isolate the gene by using restriction enzymes to cut it out
3. Make multiple copies of the gene using a PCR machine
4. Remove an embryo from an organism and separate the cells
5. Use **microinjection** to inject gene into multiple zygotes (cells)
6. Zygotes divide to form embryos
7. These embryos are then inserted into surrogate goats

Use in medical field

Transgenic goats born can produce antithrombin in their milk which is used in medicine to treat people that have issues with excessive blood clotting - reducing risk of a cardiovascular disease

Transgenic Plant

1. Identify the **Vitamin A gene in daffodils** and cut it out using restriction enzymes
2. Use a PCR machine to make multiple copies of this gene
3. Cut multiple plasmids from **soil agrobacterium T** using the same restriction enzymes
4. Paste the Vitamin A gene into the bacteria's plasmids
5. Use heat shock to insert plasmids back into soil agrobacterium T
6. Dip rice embryos in a solution of transgenic soil agrobacterium T
7. Bacteria will infect rice embryos inserting the Vitamin A gene from its plasmids into the rice embryos
8. Rice embryos will grow into rice plants that produce higher values of Vitamin A

Use in agriculture

Transgenic Bt cotton can now produce a protein that is poisonous to its main pest. Therefore the plant now requires minimal pesticides so that caterpillars are eliminated but beneficial insects are unaffected. Therefore it does not harm native animals and it is also therefore less expensive to grow with increased yield + reduces pesticide use. However it can reduce biodiversity by disrupting the food chain and outcompeting other species in the area.

Or Golden rice can now have additional nutritional value - this can be given to people in third world countries to

relieve hunger and reduce incidences of blindness
(daffodile vitamin A Gene)
To put into agrobacterium
Agrobacterium infects the rice embryos

Transgenic mosquitoes mating with malaria vectors, which eradicates the vector's ability to pass on a pathogen (CRISPR-Cas9 gene-editing technology was used to make changes in mosquito genes that could reduce their ability to spread malaria. This could provide a new way to reduce illnesses and deaths caused by malaria.

• evaluate the benefits of using genetic technologies in agricultural, medical and industrial applications

1. Define all the subjects (genetic technology)
2. Write points for and against (providing support with examples)
3. Make a judgement of how effective genetic technology is for improving human life, animal health, environment, biodiversity

Example: Modern genetic technology allows humans to manipulate the path of evolution by combining the genes of organisms that were once separate species. This has the benefit of increasing biodiversity in the short term but can decrease biodiversity if largely used. However these genetic technologies have regulations in place, have many known benefits for humans and risks are potentially not definite. Therefore the benefits outweigh the risks of using genetic technology on agriculture, medicine and industrialisation.

Agriculture

For	Against
- Potential to produce crops that are better suited to an environment (weather), disease and pest resistant and have greater output, increasing productivity e.g. BT cotton	- Potential health risks of GM products are not yet not known - People with allergies may have allergic reactions to foods they could previously eat if those foods include the DNA of other organisms they are allergic to - Bt cotton has the potential to kill all caterpillars and as a result species that feed off these caterpillars reduce in population, reducing biodiversity - Artificial insemination initially increases allele frequency desirable traits in the gene pool e.g. of dairy cows. But if these same features keep on being selected, over time this will lead to decreased overall genetic diversity of dairy cows

Medical

For	Against
- Foods with higher nutritional value may be developed to support people in third world countries - Can be used to reduce pesticide use, improving consumer health - Can be used to artificially produce medicine efficiently e.g insulin - Genes can be cloned and then used in gene therapy which involves inserting a 'healthy' gene into another organism's cells to replace damaged genes that cause disease.	- Gene therapy is still in early / experimental stages and doesn't work for all patients and side effects can lead to tumor formation

Industrial

For	Against
- GM plants could be used to produce environmentally friendly chemicals that can replace the non-renewable products that are currently used (fuels, plastics, paints) - Researchers have been working on developing genetically engineered plants	- Unknown effect on environment of releasing/using genetically modified bacteria e.g. to

and bacteria that can absorb heavy metals, such as mercury from contaminated sites. This could aid in the recovery of local ecosystems that are heavily polluted - GMO bacteria has the potential to produce enzymes that can break down oil and consume this oil, hence it can clean up oil spills in the ocean - Sugarcane is used as a biofuel where it can be fermented to produce ethanol to be used as a fuel for motor vehicles (renewable source of energy compared to oil fuels)	clean oil spills / or absorb heavy metals - Biofuels still produce large amounts of carbon dioxide into the environment, contributing to global warming
---	--

- evaluate the effect on biodiversity of using biotechnology in agriculture

Define biotechnology:

- area of biology, involving the use of living systems and organisms to develop or make products
- can have both a positive and negative impact on the genetic variation of individuals, populations, species
- reduction in genetic variation lessens the species resilience to changes in the environment and disease, as well as reducing the ability of the environment to meet the needs of humans in the future

Define a biotech

What is the biotech used for

Positive and negative effects of biotechnology

- Reproductive tech (artificial, selection, insemination & pollination)
Reduction in genetic variation due to repetitive selection of specific desired genes.
E.g. Artificial insemination has allowed for the breeding of Friesian dairy cows in Africa, resulting in greater milk output and economic gain. However this strain threatened the native ankole cattle as they can outcompete native species. This is a concern as ankle cattle are well adapted to the semi arid condition of countries such as uganda and do not require great amount of water, feed or veterinary attention and therefore are well adapted to the environment
- Artificial insemination and pollination can have advantages as it allows for the introduction of new genes to a population, which can increase the genetic variation in the gene pool.
- Whole organism cloning poses a risk to biodiversity with the potential for multiple identical genetic copies becoming widespread in a population resulting in a loss of natural vigour. Transgenic and genetically modified species pose a significant threat to genetic variation, with threats to agriculture and the natural environment as a result of selection of a narrow range of transgenic over wild species, risk of pesticide and insect resistant species leading to invasive pests in the natural environment, reduction of non-target species such as honeybees as a result of GM
- Whole organism cloning and genetic cloning has advantages for biodiversity as holds the potential to reintroduce lost genetic variation into a population restore ecosystem function as well as genetic resilience. Can also be used to increase populations of species going extinct
- Can also decrease environmental impact agriculture has on the environment (BT cotton)

Make a judgement

- Biotechnology has led to a reduction in biodiversity
- However, it should be used as it is vital to provide enough food for the growing population and can be used responsibly to maintain and even increase biodiversity

- To continue safe use of biotechnology we must implement ethical guidelines, biosafety protocols and clear legislations for genetic and animal manipulation to prevent damage to biodiversity

E.g. GMOs - T - cotton effect on biodiversity

- Introduce new alleles into a population, increasing genetic diversity - GM cotton results in reduced use of pesticides = reduced harm on surrounding plants and animals = increasing in biodiversity	- Ongoing selection of this variety results in loss of other varieties, reducing genetic diversity - GM cotton reduces specie diversity by reducing caterpillar population and other organisms that feed off caterpillars - Can outcompete other variants, resulting in loss of genetic diversity
---	---

- interpret a range of secondary sources to assess the influence of social, economic and cultural contexts on a range of biotechnologies

Issue: should biotechnology be used?

Why biotechnology is used / should be used	Why biotechnology is not used / should not be used
Social	
- Has allowed the medical industry to improve through engineering genes to rid disease causing agents and mutations. This is also present in food with biotechnology engineering food/diets to treat chronic diseases - Agricultural biotech allows us to increase supply to meet the demands of the ever growing population - Can create super foods that have higher levels of vitamins, iron, antioxidants to those who are malnourished, helping developing countries to avoid anemia, blindness and other developmental issues that stem from undernutrition and deficiencies - May help employ sustainable farming practices to combat climate change and environmental damage - Transgenic crops don't use pesticide → heightens quality of life through less use of chemicals improving human and ecological health. - Transgenic crops are drought resistant → water conservation - environmental restoration and the management of endangered species and ecosystems; for example, the collection of coral sperm and eggs at Heron Island to help regenerate coral reef	- Countries with health issues that do not affect the rest of the world (poor countries) have little influence to investigating biotechnological solutions E.g. ebola vaccines have not been developed as they only plague developing countries - Lack of consistent regulations internationally restricts GM foods to be imported - concerns over food security & political freedom → biotechnology may mean farmers have less say in the crops produced, & food production may be controlled by the government - there is a need for regulations & labelling to ensure consumer equity & knowledge, and until these regulations are effectively implemented, biotechnology may not be employed - ethical concerns about the exploitation of animals → people do not want to use biotechnology if it harms animals - potential for environmental damage via an increase in monocultures, loss of habitats & biodiversity. Eg. use of insecticide-containing GMOs impact non-harmful insects & the food chain. If herbicide-tolerant genes escape into weed populations, there is potential for weed problems to escalate.
Economic	
- Can increase efficiency of farming, pharmaceuticals and industrialisation, decreasing costs to consumers, increasing profits made by companies	- Economies with rich biodiversity may want to protect against the use of biotechnology to maintain the high rates of gene frequencies, therefore reducing the use

- Economies with rich biodiversity are more likely to have specie varieties with advantageous products that can be useful in the medical field, improving quality of life E.g. The Philippines has rich biodiversity that has led to the third largest pharmaceutical market in the world. - May provide farmers in third world countries with tools to grow crops easily and quickly - Enables farmers with nutrient-poor soil or poor access to water to continue growing nutrient-rich foods, and increasing foot output - Can reduce poverty in developing countries through growing of GMOs that are more efficient and therefore more affordable E.g. Golden rice to feed malnourished in Africa	of biotechnology E.g. Philippines banned the import of GM soybeans - Large companies often hold the license to grow their developed GM products. Therefore this decentifies small farmers from using this biotechnology as it reduces profits due to paying rights to grow these products - Potential for monopolisation by large biotechnology companies, cutting small scale farmers out of the global market in third world countries, creating a greater gap in income inequality (socioeconomic disparity) E.g. Developing countries traditionally export resources such as palm oil, vanilla and rubber. But without GM products they may lose their international competitiveness in the market
Cultural	
- Food is an important element of different cultures. Therefore different societies may be inclined to use biotechnology to preserve their cultures signature dish and maintain significant industries in certain areas E.g. Taro is sacred in the Hawaiian culture but over the last century taro has been reduced due to taro leaf blight disease but GM taro has the potential to be disease resistant - Biotechnology can be used to adapt to changing environments - Less environmental damage because of efficient food production. This is because some indigenous cultures have sacred land E.g. golden rice could be produced instead of normal rice which takes up less room as it has higher nutritional value and therefore damages less land	- Traditional, region-specific farming practices may be eradicated in favour of large-scale agricultural methods E.g. loss of important parts of indigenous cultures - Cultural concerns that GM taro may reduce the purity of the taro, alter the food's identity, reduce the biodiversity of taro, removing the option to grow different variants of taro - Backlash from conservative \ certain religious groups on ethical grounds. E.g. in isalm eating pork is prohibited, therefore there are concerns of pig genes into food sources or in organ transplants, no longer allowing them to use these foods